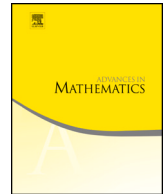




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Marked length spectrum rigidity for Anosov magnetic surfaces



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ABSTRACT

We show that if M is a closed, connected, oriented surface, and two Anosov magnetic systems on M are conjugate by a volume-preserving conjugacy isotopic to the identity, with their magnetic forms in the same cohomology class, then the metrics are isometric. This extends the recent result by Guillarmou, Lefeuvre, and Paternain to the magnetic setting.

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1. Introduction

Let M be a closed, connected, oriented manifold, and let g be a Riemannian metric on M . Denoting the geodesic flow on its unit tangent bundle by $\varphi_g^t : S_g M \rightarrow S_g M$ and denoting the infinitesimal generator of the flow by X_g , we say that the metric g is *Anosov* if φ_g^t is an *Anosov flow*, i.e., there exists a dense orbit and a φ_g^t -invariant splitting $TS_g M = E^+ \oplus \mathbb{R}X_g \oplus E^-$ and constants $c, d > 0$ such that for all $t \geq 0$ and $\xi \in E^\pm(v)$, we have

$$\|d_v \varphi_g^{\mp t}(\xi)\| \leq de^{-ct}\|\xi\|, \quad (1.1)$$

where $\|\cdot\|$ denotes the norm coming from the Sasaki metric on $S_g M$.

Anosov metrics can be seen as a generalization of negatively curved metrics [1] and, as such, they share many common qualities. For example, denoting the collection of free homotopy classes by $\mathcal{C}$, observe that if g is an Anosov metric, then for every non-trivial free homotopy class $\nu \in \mathcal{C}$ there is a unique closed g -geodesic $\gamma_\nu \in \nu$ [32]. The *marked length spectrum* is defined to be the function $\text{MLS}(g) : \mathcal{C} \rightarrow [0, \infty)$ which assigns to a non-trivial free homotopy class ν the length of its geodesic representative γ_ν , and 0 to the trivial free homotopy class. It is conjectured that $\text{MLS}(g)$ uniquely characterizes the Anosov metric g . More precisely:

Conjecture 1.1 ([27]). *Let g_1 and g_2 be Anosov metrics on M . If $\text{MLS}(g_1) = \text{MLS}(g_2)$, then g_1 and g_2 are isometric by an isometry which is isotopic to the identity.*

In the case where g_1 and g_2 are both negatively curved metrics, this is the well-known *Burns-Katok conjecture* [7]. Using Livshits' theorem [33] along with an argument by Gromov [25], one can show that if $\text{MLS}(g_1) = \text{MLS}(g_2)$, then there is a C^0 -conjugacy between the corresponding geodesic flows which is isotopic to the identity. With this in mind, the conjecture broadly states that dynamical invariants give rise to geometric rigidity. It is known that Conjecture 1.1 holds for non-positively curved metrics on surfaces [11,13,40], and for negatively curved metrics in dimensions three and higher when one of the metrics is locally symmetric [6,28]. For Anosov metrics, the conjecture has been recently proved for surfaces in [27]; a local version (i.e. when the metrics are sufficiently close in an appropriate topology) holds in arbitrary dimension, provided the metrics have non-positive sectional curvature [26]. The conjecture is still open in general.

In this paper, we bring the above discussion to the context of magnetic dynamics. Namely, we study marked length spectrum rigidity for Riemannian metrics on surfaces in presence of a magnetic field. Magnetic dynamics was formalized in [2,3], and has been investigated in many works in the last decades. For a general overview, we refer the reader to [4] and the references therein.

From now on, let M be a closed, connected, orientable surface. A *magnetic system* is a pair (g, b) , with g a Riemannian metric on M and b a smooth function (referred to as

the *magnetic intensity*) on M . Denote by Ω_g the area form induced by g on M , and let i be the complex structure given by the orientation of Ω_g . More precisely, given a vector $v \in TM$, iv is the rotation of v by the angle $\pi/2$ according to the orientation. A curve $\gamma : \mathbb{R} \rightarrow M$ is a (g, b) -geodesic if it satisfies the differential equation

$$\frac{D\dot{\gamma}}{dt}(t) = b(\gamma(t))i\dot{\gamma}(t), \quad (1.2)$$

where D/dt denotes the covariant derivative along γ induced by the Levi-Civita connection of g . A (g, b) -geodesic describes the motion of a charged particle on (M, g) under the influence of a magnetic force described by b .

Observe that if $b = 0$, then (1.2) reduces to the standard equation for g -geodesics on M . Also observe that while (g, b) -geodesics have constant speed, they are not homogeneous if $b \neq 0$. In particular, rescaling the initial speed vector or inverting the direction of the initial velocity vector may drastically change the associated trajectory. Despite this, up to a time reparameterization, one can recover the dynamics of unit speed (g, b) -geodesics from the dynamics of (g, b) -geodesics with speed $s > 0$ by rescaling the metric g by the factor $1/s^2$ and the magnetic intensity by the factor $1/s$. Without loss of generality, we restrict our perspective to unit speed solutions of (1.2).

The *magnetic flow* associated to the pair (g, b) is the flow $\varphi_{g,b}^t : S_g M \rightarrow S_g M$ given by $\varphi_{g,b}^t(\gamma(0), \dot{\gamma}(0)) := (\gamma(t), \dot{\gamma}(t))$. The orbits of $\varphi_{g,b}^t$ have a geometric interpretation in terms of the geodesic curvature. The *g-geodesic curvature* associated to a curve $\gamma : \mathbb{R} \rightarrow M$ is defined by

$$k^g(t) := g\left(\frac{D\dot{\gamma}}{dt}(t), i\dot{\gamma}(t)\right). \quad (1.3)$$

It is immediate from (1.2) that a unit speed curve γ is a (g, b) -geodesic if and only if the g -geodesic curvature of γ satisfies $k^g(t) = b(\gamma(t))$.

The magnetic system (g, b) is called an *Anosov magnetic system* if the flow $\varphi_{g,b}^t$ is Anosov. Similar to the geodesic scenario, the existence of an Anosov magnetic system on M implies some restrictions on both the magnetic intensity and the surface. By [37, Corollary C], the genus of M must be at least two and, as pointed out in [8, Theorem B], the average of the magnetic intensity cannot be larger than the so-called *Mañé critical value*.

An interesting problem is determining how to construct Anosov magnetic systems. By the structural stability of Anosov flows, observe that if g is an Anosov metric, then (g, b) is Anosov as long as b is uniformly small. On the other hand, not every Anosov magnetic system comes from an Anosov metric: in [8, Section 7], it was shown that one can start with a non-Anosov metric on a higher genus surface and place a magnetic intensity b on M so that (g, b) is Anosov. This shows that the set of metrics for which there is a magnetic intensity b so that (g, b) is Anosov is larger than the set of Anosov metrics.

As in the Riemannian setting, an important subset of the set of Anosov magnetic systems consists of those which are negatively curved in a magnetic sense. The *magnetic curvature* associated to the system (g, b) is given by

$$\mathbb{K}^{g,b} : S_g M \rightarrow \mathbb{R} \quad \mathbb{K}^{g,b}(x, v) := K^g(x) - d_x b(iv) + b^2(x), \quad (1.4)$$

where K^g is the Gaussian curvature of (M, g) . It was shown in [46] that if $\mathbb{K}^{g,b} < 0$, then the magnetic flow is Anosov.¹ Under additional assumptions, this result has been extended to non-positively curved magnetic flows in the recent paper [37] of the third author in collaboration with Yumin Shen.

Given an Anosov magnetic system (g, b) , it is still true that for each non-trivial free homotopy class, there is a unique closed (g, b) -geodesic with unit speed [10]. We define the *magnetic marked length spectrum* to be the function $\text{MLS}(g, b) : \mathcal{C} \rightarrow [0, +\infty)$ which associates to a non-trivial free homotopy class ν the length of the unique closed unit speed (g, b) -geodesic γ_ν in ν , and zero to the trivial free homotopy class.

Remark 1.2. Observe that in the Riemannian case, the traces of γ_ν and $\gamma_{-\nu}$ always coincide. This is not necessarily the case for magnetic geodesics. In particular, for some ν , it may be that $\text{MLS}(g, b)(\nu) \neq \text{MLS}(g, b)(-\nu)$, although there are non-trivial examples where there is equality for every choice of ν (e.g. Lemma 4.1).

Our main result shows that the magnetic marked length spectrum, the area, and the cohomology class of $[b\Omega_g]$ determine the Riemannian metric and the magnetic intensity, extending the result of the third author in [35,36] from the deformative setting to the global setting.

Theorem 1.3. *Let M be a closed, connected, oriented surface. Suppose (g_1, b_1) and (g_2, b_2) are Anosov magnetic systems on M satisfying the following conditions:*

$$\text{MLS}(g_1, b_1) = \text{MLS}(g_2, b_2), \quad \text{Area}(g_1) = \text{Area}(g_2), \quad \text{and} \quad [b_1\Omega_{g_1}] = [b_2\Omega_{g_2}].$$

Then there exists a diffeomorphism $f : M \rightarrow M$ which is isotopic to the identity and which satisfies $f^(g_2) = g_1$ and $f^*(b_2) = b_1$.*

It follows from Livshits' theorem [33] along with a theorem of Ghys [20] that equality of the marked length spectrum implies that there is a C^0 -conjugacy between the corresponding magnetic flows which is isotopic to the identity. In the case where the magnetic intensities are trivial, one can use the work of Feldman-Ornstein [19] to deduce that the conjugacy is smooth. The recent rigidity result by Gogolev-Rodriguez Hertz in [22, Theorem 1.1] extends this work to a larger class of Anosov flows on three manifolds,

¹ See also [23,24] for earlier works in this direction, and [4, Appendix A] for the proof in higher dimensions.

and, in particular, shows that equality of the magnetic marked length spectrum implies that the flows φ_{g_1, b_1}^t and φ_{g_2, b_2}^t are C^∞ -conjugate by some $h : S_{g_1}M \rightarrow S_{g_2}M$ which is isotopic to the identity (cf. [45]).

Remark 1.4. Using the positive proportion Livshits theorem proven by Dilsavor and the third author in [17], one can relax the assumption that the marked length spectra agree on all free homotopy classes to the assumption that the marked length spectra agree on a set of free homotopy classes which has positive asymptotic upper density with respect to length. This can be interpreted as saying that one only needs to verify a positive percentage of the marked length spectra agree in Theorem 1.3.

Let α^g be the contact form associated to the geodesic flow φ_g^t . Recall that the *Liouville volume form* on S_gM associated to the metric g is the top-degree form $\mu^g := -\alpha^g \wedge d\alpha^g$. As we will see in Lemma 2.1, one can use the area assumption to show that the conjugacy described above must pullback the Liouville volume form for g_2 to the Liouville volume form of g_1 , and thus we can reduce Theorem 1.3 to the following.

Theorem 1.5. *Let M be a closed, connected, oriented surface. Suppose (g_1, b_1) and (g_2, b_2) are Anosov magnetic systems on M such that the corresponding magnetic flows are smoothly conjugate by a conjugacy h which is isotopic to the identity, the conjugacy h preserves the corresponding Liouville volume forms, and $[b_1\Omega_{g_1}] = [b_2\Omega_{g_2}]$. Then there is a diffeomorphism $f : M \rightarrow M$ which is isotopic to the identity and which satisfies $f^*(g_2) = g_1$ and $f^*(b_2) = b_1$.*

The proof of Theorem 1.5 follows a similar scheme as the one in [27]. Namely, we want to reduce to the case where the metrics are conformally equivalent, and then we want to apply an argument similar to the one found in [31]. As discussed above, we are able to reduce to the case where the magnetic flows are smoothly conjugated. Once we have that this conjugacy is smooth, a recent result by Echevarría Cuesta shows that the metrics have to be conformally related [14]. However, this is where the arguments between the geodesic and magnetic cases diverge.

In the geodesic setting, one can take advantage of equidistribution and the fact that geodesics are length minimizers in order to deduce that the metrics have to be isometric. While magnetic geodesics can be realized as critical points of a functional, this functional does not respect necessarily the periods like in the Riemannian setting. In order to get something which resembles Katok's arguments, we have to take advantage of the fact that the magnetic flow is homologically full as well as the fact that the cohomology classes match in order to show that the length functional is minimized on magnetic geodesics up to an error of a 1-form. Using the fact that the Liouville measure is flip invariant and equidistribution of homologically trivial orbits, we show that this is enough to conclude that the metrics are isometric and the magnetic intensities must match.

We note that the assumptions on the area and the cohomology class in Theorem 1.3 are necessary via two examples in Section 4, which can also be found in [41]. For the

area assumption, we show in Lemma 4.1 that for a hyperbolic metric g and a constant $|b| < 1$, we have $\text{MLS}(g, b) = (1 - b^2)^{-1/2} \text{MLS}(g)$. By considering the homothetic metric $g_b := (1 - b^2)^{-1}$, we see that $\text{MLS}(g, b) = \text{MLS}(g_b)$ while $\text{Area}(g) \neq \text{Area}(g_b)$. It follows that the marked length spectrum assumption alone is not enough to guarantee that the underlying metrics are isometric.

It is shown in Lemma 2.2 that $[b_1 \Omega_{g_1}] = \pm [b_2 \Omega_{g_2}]$ (see also [41, Lemma 4.1] and [14, Proposition 4.5]). Lemma 4.2 shows that this is actually realizable, in the sense that for a hyperbolic metric g and a constant $|b| < 1$, the magnetic systems (g, b) and $(g, -b)$ are smoothly conjugate by a conjugacy which is isotopic to the identity. In particular, this shows that the marked length spectrum assumption alone is not sufficient to guarantee that the cohomology classes of $b_1 \Omega_{g_1}$ and $b_2 \Omega_{g_2}$ are the same.

We now give some applications of Theorem 1.3. First, we observe that Theorem 1.3 extends [27, Theorem 1.1] to the case of *exact magnetic systems*, i.e., magnetic systems where $[b\Omega_g] = 0$. Using Lemma 2.2, it can be shown that if (g_1, b_1) and (g_2, b_2) are exact Anosov magnetic systems which are smoothly conjugate, then $\text{Area}(g_1) = \text{Area}(g_2)$. In fact, this also extends to $(g_1, b_1 + \delta)$ and $(g_2, b_2 + \delta)$, provided that $\delta \in \mathbb{R}$ is sufficiently small so that the marked length spectrum still makes sense. With this in mind, the following is immediate.

Corollary 1.6. *Let M be a closed, connected, orientable surface, and let (g_1, b_1) and (g_2, b_2) be exact Anosov magnetic systems on M . If $\delta \in \mathbb{R}$ is sufficiently small so that $(g_1, b_1 + \delta)$ and $(g_2, b_2 + \delta)$ are still Anosov and $\text{MLS}(g_1, b_1 + \delta) = \text{MLS}(g_2, b_2 + \delta)$, then there exists a diffeomorphism $f : M \rightarrow M$ which is isotopic to the identity and which satisfies $f^*(g_2) = g_1$ and $b_2 \circ f = b_1$.*

Notice that there is no assumption on the metric in Corollary 1.6 – as long as the exact magnetic systems $(g_i, b_i + \delta)$ share the same marked length spectrum, then the underlying metrics are isometric. As previously mentioned, notice that $\text{MLS}(g, b)$ encodes the length of the unique closed curve γ in each non-trivial free homotopy class of M with prescribed g -geodesic curvature equal to $b \circ \gamma$. The above result can then be interpreted as saying that the unit speed curves with g -geodesic curvature equal to b completely determine the underlying metric, provided the set of these curves is “dynamically rich.” In Section 4, we modify the example in [8, Section 7] to construct an example of a metric g which is not Anosov, but for which there exists a magnetic intensity b so that $[b\Omega_g] = 0$ and the magnetic system (g, b) is Anosov. Thus, Corollary 1.6 along with this example shows that marked length spectrum rigidity can be extended beyond the setting of Anosov metrics and non-positively curved metrics.

Next, we see how the magnetic marked length spectrum uniquely characterizes geodesic flows, provided the area is fixed. Using the arguments in [24, Théorème 7.3], one can show that if (g_1, b) and g_2 both satisfy $\mathbb{K}^{g_1, b}, K^{g_2} < 0$, $\text{MLS}(g_1, b) = \text{MLS}(g_2)$, and $\text{Area}(g_1) = \text{Area}(g_2)$, then $b \equiv 0$ and g_1 and g_2 are isometric. One can also deduce from Lemma 2.2 that if $\text{MLS}(g_1, b) = \text{MLS}(g_2)$ and $\text{Area}(g_1) = \text{Area}(g_2)$, then $[b\Omega_g] = 0$. This

shows that Corollary 1.6 extends [24, Théorème 7.3] from negatively curved magnetic systems to Anosov magnetic systems.

Corollary 1.7. *Let (g_1, b) be an Anosov magnetic system. If g_2 is an Anosov metric such that $\text{MLS}(g_1, b) = \text{MLS}(g_2)$ and $\text{Area}(g_1) = \text{Area}(g_2)$, then $b \equiv 0$ and there is an isometry between g_1 and g_2 which is isotopic to the identity.*

We also observe a connection to another problem in dynamical systems. Given a magnetic system (g, b) , we denote the space of probability measures on $S_g M$ which are invariant under the magnetic flow by $\mathcal{M}(g, b)$, and for $\mu \in \mathcal{M}(g, b)$ we denote the metric entropy by $h_\mu(g, b)$. By the variational principle, the topological entropy of the magnetic flow can be expressed as

$$h_{\text{top}}(g, b) = \sup_{\mu \in \mathcal{M}(g, b)} h_\mu(g, b).$$

Katok's entropy conjecture in the context of magnetic flows says that one has the relationship $h_\mu(g, b) = h_{\text{top}}(g, b)$ if and only if the magnetic flow is “algebraic” (see [16, Conjecture 1.1 and Question 1] for the more general statement of Katok's entropy conjecture). In the case where $b \equiv 0$, this translates to the metric being a hyperbolic metric, and this was established in [30]. A consequence of the Corollaries 1.6 and 1.7 along with [16, Theorem A] shows that something similar happens in the case of exact Anosov magnetic systems.

Corollary 1.8. *Let (g, b) be an exact Anosov magnetic system. If ν^g denotes the normalized Liouville volume measure, then $h_{\nu^g}(g, b) = h_{\text{top}}(g, b)$ if and only if $b \equiv 0$ and g is a hyperbolic metric.*

Remark 1.9. More generally, if (g, b) is an Anosov magnetic system, then one has $h_{\nu^g}(g, b) = h_{\text{top}}(g, b)$ if and only if the weak stable and unstable bundles are smooth [29]. One can then use [42, Corollary 1] to conclude that g must be hyperbolic and b constant. Combined with [16, Theorem A], we have that $\text{MLS}(g, b) = \text{MLS}(g_0)$ for some hyperbolic metric g_0 . However, as noted in Section 4, this is insufficient for concluding that $b \equiv 0$; one also needs an area assumption.

Finally, we introduce the magnetic marked length spectrum rigidity problem in higher dimensions. Let M be a closed, connected manifold of arbitrary dimension. A *magnetic system* is a pair (g, σ) , where g is a Riemannian metric and σ is a closed 2-form (referred to as the *magnetic form*). Note that the magnetic form generalizes the magnetic intensity b introduced for surfaces: if σ is a closed 2-form on a surface, then the magnetic intensity is precisely the unique smooth function b on M for which $\sigma = b\Omega_g$. A (g, σ) -*geodesic* is a curve $\gamma : \mathbb{R} \rightarrow M$ which satisfies the equation

$$\frac{D\dot{\gamma}}{dt}(t) = Y(\dot{\gamma}(t)), \quad (1.5)$$

where $Y : TM \rightarrow TM$ is the *Lorentz operator* defined by $g(Y(v), w) = \sigma(v, w)$ for all $v, w \in TM$. As in the surface case, solutions of (1.5) have constant speed, and the *magnetic flow* $\varphi_{g,\sigma}^t : S_g M \rightarrow S_g M$ associated to (g, σ) is the flow on $S_g M$ induced by (1.5). A pair (g, σ) is *Anosov* if the flow $\varphi_{g,\sigma}^t$ is Anosov, and as before the magnetic marked length spectrum naturally generalises to Anosov pairs (g, σ) . With this in mind, a natural question is the following.

Question 1.10. *Let (g_1, σ_1) and (g_2, σ_2) be two Anosov magnetic systems on a closed, connected, oriented manifold M . If $\text{MLS}(g_1, \sigma_1) = \text{MLS}(g_2, \sigma_2)$, $\text{Vol}(g_1) = \text{Vol}(g_2)$, and $[\sigma_1] = [\sigma_2]$, is it true that g_1 and g_2 must be isometric?*

As evidence towards a positive answer for this question, suppose that g_1 and g_2 are two Anosov metrics with non-positive sectional curvature. If (g_1, σ_1) is an Anosov magnetic system satisfying $\text{MLS}(g_1, \sigma_1) = \text{MLS}(g_2)$ and $\text{Vol}(g_1) = \text{Vol}(g_2)$, then it follows from the fact that geodesics are the unique length minimizers in their free homotopy class that $\text{MLS}(g_1) \leq \text{MLS}(g_2)$. As long as the metrics are sufficiently close according to [26, Theorem 2], we have that g_1 and g_2 are isometric and $\sigma_1 \equiv 0$.²

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2. Preliminaries

Throughout, let M be a closed, connected, oriented surface, let (g, b) be a magnetic system on M , and let $\pi_g : S_g M \rightarrow M$ be the footprint map for the unit tangent bundle. The *geodesic vector field* X_g is the infinitesimal generator of the geodesic flow φ_g^t on $S_g M$. Since M is oriented, we may consider the *rotation flow* $\rho_g^t : S_g M \rightarrow S_g M$, given by $\rho_g^t(x, v) := (x, e^{it}v)$. Its infinitesimal generator is the *vertical vector field*, which we denote by V_g . Finally, the *horizontal vector field* on $S_g M$ is the vector field given by $H_g := [V_g, X_g]$. Observe that the vector fields X_g, H_g , and V_g form a global frame on $S_g M$, which we refer to as *Cartan's moving frame*. They satisfy the relations

$$[V_g, H_g] = -X_g, \quad [V_g, X_g] = H_g, \quad [X_g, H_g] = \pi_g^*(K^g)V_g, \quad (2.1)$$

² Note that one can also use this argument along with [12, Theorem 1.1] to deduce Corollary 1.7 in the case where g_1 and g_2 are negatively curved metrics.

where K^g is the Gaussian curvature on M . Dual to these vector fields are the 1-forms α^g, β^g , and ψ^g on $S_g M$, which we refer to as *Cartan's moving coframe*. Using (2.1), one can show that they satisfy

$$d\alpha^g = \psi^g \wedge \beta^g, \quad d\beta^g = -\psi^g \wedge \alpha^g, \quad d\psi^g = -\pi_g^*(K^g)\alpha^g \wedge \beta^g. \quad (2.2)$$

Observe $\pi_g^*(\Omega_g) = \alpha^g \wedge \beta^g$, hence we can rewrite the last relation as

$$d\psi^g = -\pi_g^*(K^g\Omega_g). \quad (2.3)$$

Moreover, the Liouville volume form μ^g satisfies $\mu^g = \alpha^g \wedge \beta^g \wedge \psi^g$.

Next, observe that the magnetic flow $\varphi_{g,b}^t$ is a Hamiltonian flow, induced by the function $H : TM \rightarrow \mathbb{R}$ given by $H(x, v) := \frac{1}{2}g_x(v, v)$ and the symplectic form

$$\omega_{g,b} := -d\alpha_g + \pi_g^*(b\Omega_g). \quad (2.4)$$

With this, we now have the tools to prove the following.

Lemma 2.1. *Let (g_1, b_1) and (g_2, b_2) be Anosov magnetic systems on M , smoothly conjugated by $h : S_{g_1} M \rightarrow S_{g_2} M$; then:*

$$h^*(\mu^{g_2}) = \frac{\text{Area}(g_2)}{\text{Area}(g_1)} \mu^{g_1}.$$

Proof. Notice that there exists $J_h \in C^\infty(S_{g_1} M)$ so that $h^*(\mu^{g_2}) = J_h \mu^{g_1}$. Since the magnetic flow is Hamiltonian, it preserves the Liouville volume form, and thus

$$J_h \mu^{g_1} = h^*(\mu^{g_2}) = h^*((\varphi_{g_2, b_2}^t)^*(\mu^{g_2})) = (\varphi_{g_1, b_1}^t)^*(h^*(\mu^{g_2})) = (J_h \circ \varphi_{g_1, b_1}^t) \mu^{g_1}.$$

We deduce that J_h is constant along orbits. Since the magnetic flow is Anosov, it admits a dense orbit, and thus J_h must be constant everywhere. Next, recall that

$$\int_{S_{g_i} M} \mu^{g_i} = 2\pi \text{Area}(g_i).$$

Using a change of variables, we have

$$J_h \int_{S_{g_1} M} \mu^{g_1} = \int_{S_{g_1} M} h^*(\mu^{g_2}) = \int_{S_{g_2} M} \mu^{g_2}.$$

Thus, J_h is the ratio of the areas. $\square$

Using the Hamiltonian property, one can deduce that the infinitesimal generator of the magnetic flow is given by $X_{g,b} = X_g + \pi_g^*(b)V_g$, implying that

$$\iota_{X_{g,b}}\mu^g = \omega_{g,b}. \quad (2.5)$$

Next, since $b\Omega_g$ is a closed 2-form on M , we may use the isomorphism $H^2(M, \mathbb{R}) \cong \mathbb{R}$ to get

$$b\Omega_g = c_{g,b}K^g\Omega_g + d\theta, \quad (2.6)$$

where θ is a 1-form on M and

$$c_{g,b} := \frac{1}{2\pi\chi(M)} \int_M b\Omega_g. \quad (2.7)$$

In particular, we see that $\pi_g^*(b\Omega_g) = d(-c_{g,b}\psi_g + \pi_g^*(\theta))$, and thus if

$$\tau_{g,b} := -\alpha_g - c_{g,b}\psi_g + \pi_g^*(\theta), \quad (2.8)$$

then $\omega_{g,b} = d\tau_{g,b}$ on S_gM . Using Equation (2.5) and [9, Lemma 7.1], this shows that the magnetic flow is *homologically full*, meaning that every integral homology class on S_gM has a closed (g, b) -geodesic representative $(\gamma, \dot{\gamma})$.³ It is well-known that this implies that for all closed 1-forms η on $S_{g_1}M$, we have

$$\int_{S_{g_1}M} \eta(X_{g_1,b_1})\mu^{g_1} = 0; \quad (2.9)$$

see, for instance, [44] and [21, Theorem 2.4]. Furthermore, this primitive is a useful tool in showing that the existence of a smooth volume preserving conjugacy implies that $[b_1\Omega_{g_1}] = \pm[b_2\Omega_{g_2}]$.

Lemma 2.2 ([14, Proposition 4.5], [41, Lemma 4.1]). *Let (g_1, b_1) and (g_2, b_2) be Anosov magnetic systems on M . If they are smoothly conjugated, then*

$$A_1 \left[1 + \frac{2\pi\chi(M)c_{g_2,b_2}^2}{A_2} \right] = A_2 \left[1 + \frac{2\pi\chi(M)c_{g_1,b_1}^2}{A_1} \right],$$

where $A_i := \text{Area}(g_i)$. In particular, if $A_1 = A_2$, then $c_{g_1,b_1} = \pm c_{g_2,b_2}$.

Proof. As before, let $h : S_{g_1}M \rightarrow S_{g_2}M$ denote a smooth conjugacy between the magnetic flows. Lemma 2.1 and (2.5) imply that $h^*(\omega_{g_2,b_2}) = (A_2/A_1)\omega_{g_1,b_1}$, hence

³ One can also use [20] to deduce this result.

$h^*(\tau_{g_2, b_2}) - (A_2/A_1)\tau_{g_1, b_1}$ is a closed 1-form on $S_{g_1}M$. Thus, using (2.9), Lemma 2.1, and a change of variables formula, we may rearrange

$$\int_{S_{g_1}M} \iota_{X_{g_1, b_1}} \left(h^*(\tau_{g_2, b_2}) - \frac{A_2}{A_1} \tau_{g_1, b_1} \right) \mu^{g_1} = 0$$

into

$$\frac{A_1}{A_2} \int_{S_{g_2}M} \tau_{g_2, b_2}(X_{g_2, b_2}) \mu^{g_2} = \frac{A_2}{A_1} \int_{S_{g_1}M} \tau_{g_1, b_1}(X_{g_1, b_1}) \mu^{g_1}.$$

Let θ_i be 1-forms on M satisfying (2.8) for $b_i\Omega_{g_i}$. Notice

$$\tau_{g_i, b_i}(X_{g_i, b_i}) = -1 - c_{g_i, b_i} \pi_{g_i}^*(b_i) + \theta_i,$$

here viewing θ_i as smooth functions on $S_{g_i}M$, and thus

$$\frac{A_1}{A_2} \left[-A_2 - c_{g_2, b_2} \int_M b_2 \Omega_{g_2} \right] = \frac{A_2}{A_1} \left[-A_1 - c_{g_1, b_1} \int_M b_1 \Omega_{g_1} \right].$$

Observe that the integral of θ_i vanishes thanks to flip invariance of the Liouville measure. Applying (2.7) yields the desired result. $\square$

Remark 2.3. Note that the *helicity* of the magnetic flow $X_{g, b}$ can be calculated as

$$\mathcal{H}(X_{g, b}) := \int_{S_g M} \tau_{g, b} \wedge d\tau_{g, b} = \int_{S_g M} \tau_{g, b}(X_{g, b}) \mu^g,$$

see [9, 43]. With this in mind, we have shown the following relationship between the helicities of the magnetic flows given by (g_1, b_1) and (g_2, b_2) assuming they are smoothly conjugated:

$$\frac{\mathcal{H}(X_{g_2, b_2})}{A_2^2} = \frac{\mathcal{H}(X_{g_1, b_1})}{A_1^2}.$$

This primitive is also a useful tool for establishing the following relation.

Lemma 2.4. *Let (g_1, b_1) and (g_2, b_2) be Anosov magnetic systems on M which satisfy $\text{Area}(g_1) = \text{Area}(g_2)$ and are smoothly conjugate via the map $h : S_{g_1}M \rightarrow S_{g_2}M$. There is a closed 1-form ρ on M such that if γ_1 is a closed orbit for (g_1, b_1) and γ_2 is the corresponding closed orbit for (g_2, b_2) under h , then we have*

$$-c_{g_2, b_2} \int_{\gamma_2} b_2 + \int_{\gamma_2} \theta_2 = -c_{g_1, b_1} \int_{\gamma_1} b_1 + \int_{\gamma_1} [\theta_1 + \rho].$$

Proof. Using the Gysin sequence, $\pi_{g_1}^* : H^1(M, \mathbb{R}) \rightarrow H^1(S_{g_1}M, \mathbb{R})$ is an isomorphism [39, Corollary 8.10]. As a consequence, there is a smooth function $u : S_{g_1}M \rightarrow \mathbb{R}$ and a closed 1-form ρ on M so that

$$h^*(\tau_{g_2, b_2}) - \tau_{g_1, b_1} = \pi_{g_1}^*(\rho) + du.$$

Contracting both sides of this equation against X_{g_1, b_1} and integrating over the orbit γ_1 , the result follows from (2.9). $\square$

Another characterization of (g, b) -geodesics is that they are minimizers of an action functional. Recall if $[b\Omega_g] = 0$, so $b\Omega_g = d\theta$ for some 1-form on M , then one can define the *magnetic Lagrangian* by

$$\Lambda_{g,b}(x, v) := \frac{1}{2}\|v\|_x^2 - \theta_x(v),$$

and (g, b) -geodesics are precisely the extremals for the action functional associated to $\Lambda_{g,b}$ [8]. If $[b\Omega_g] \neq 0$ and (g, b) is Anosov, the lift of $b\Omega_g$ to the universal cover is exact, and one can work on this level.

In fact, in the Anosov setting, there is a nice way of expressing this action functional on the level of M . Inside of every non-trivial free homotopy class $\nu \in \mathcal{C}$ there exists a unique unit-speed (g, b) -geodesic $\gamma_\nu \in \nu$ [10]. Given a closed curve $\gamma : S^1 \rightarrow M$ in the free homotopy class ν , let $\Sigma_{g,b}(\gamma)$ be a 2-chain on M with boundary given by $\gamma - \gamma_\nu$. The (unit-speed) *magnetic action* of a smooth curve $\gamma : [0, T] \rightarrow M$ is given by

$$A_{g,b}(\gamma) = \frac{1}{2} \int_0^T \|\dot{\gamma}(t)\|^2 dt + \frac{T}{2} - \int_{\Sigma_{g,b}(\gamma)} b\Omega_g,$$

see [38]. In particular, we can now view unit-speed (g, b) -geodesics as the unique minimizers of $A_{g,b}$ in every free homotopy class.

In light of the action functional, define the *magnetic length* of an arbitrary closed curve γ by

$$L_{g,b}(\gamma) := \ell_g(\gamma) - \int_{\Sigma_{g,b}(\gamma)} b\Omega_g, \quad (2.10)$$

where ℓ_g denotes the (Riemannian) length of the curve with respect to the metric g . We make two observations on the magnetic length functional. The first is that if γ is a magnetic geodesic, then $L_{g,b}(\gamma) = \ell_g(\gamma)$, since $\Sigma_{g,b}(\gamma)$ is trivial. The second observation follows from the fact that magnetic geodesics are the unique minimizers of the magnetic action functional $A_{g,b}$ in a free homotopy class.

Lemma 2.5. *Let (g, b) be an Anosov magnetic system on M . If $\gamma : S^1 \rightarrow M$ is an arbitrary closed curve inside of a free homotopy class ν , then*

$$L_{g,b}(\gamma_\nu) \leq L_{g,b}(\gamma).$$

Proof. Assume for contradiction that we have $L_{g,b}(\gamma) < L_{g,b}(\gamma_\nu)$. Notice that the magnetic length is invariant under reparameterizations, so we may assume that γ has unit speed. In this case, we get $A_{g,b}(\gamma) = L_{g,b}(\gamma)$ and $A_{g,b}(\gamma_\nu) = L_{g,b}(\gamma_\nu)$, contradicting $A_{g,b}(\gamma_\nu) \leq A_{g,b}(\gamma)$. $\square$

Let (g, b) be an Anosov magnetic system on M , let $P_{g,b}$ be the collection of closed unit speed (g, b) -geodesics, and let $P_{g,b}^0 \subseteq P_{g,b}$ be the homologically trivial closed (g, b) -geodesics. Recall that the *geometric potential* for (g, b) is the function given by

$$\lambda_{g,b} : S_g M \rightarrow \mathbb{R}, \quad \lambda_{g,b}(x, v) := -\frac{d}{dt} \Big|_{t=0} \log(\det(d_v \varphi_{g,b}^t|_{E^u(v)})).$$

It is convenient to study the action of this on the level of $P_{g,b}$, which we denote by

$$\Phi_{g,b} : P_{g,b} \rightarrow \mathbb{R}, \quad \Phi_{g,b}(\gamma) := \int_{\gamma} \lambda_{g,b}.$$

For $T, \Delta > 0$, let $P_{g,b}^0(T, \Delta)$ be the collection of closed homologically trivial (g, b) -geodesics with length in the interval $[T, T + \Delta)$, define

$$N^{g,b}(T, \Delta) := \sum_{\gamma \in P_{g,b}^0(T, \Delta)} \exp(\Phi_{g,b}(\gamma)),$$

and consider the family of measures

$$\nu_{T,\Delta}^{g,b} := \frac{1}{N^{g,b}(T, \Delta)} \sum_{\gamma \in P_{g,b}^0(T, \Delta)} \exp(\Phi_{g,b}(\gamma)) \delta_\gamma,$$

where δ_γ is the probability measure supported on γ . The following homologically trivial equidistribution result was established by Coles and Sharp in [9].

Theorem 2.6 ([9, Theorem 4.1]). *If (g, b) is an Anosov magnetic system and $\Delta > 0$ is fixed, then the measures $\nu_{T,\Delta}^{g,b}$ converge to the normalized Liouville volume measure ν^g in the weak* topology as $T \rightarrow \infty$.*

Finally, it will be convenient to work on the projective tangent bundle, which is independent of the choice of metric. Namely, let SM be the principal circle bundle over M with $p : SM \rightarrow M$ the projection map and fibers $S_x M = (T_x M \setminus \{0\}) / \sim$, where $v \sim w$ if and only if $v = Cw$ with $C > 0$. Given any Riemannian metric g on M , we can

identify $S_g M$ with SM by sending vectors to their equivalence classes. When working on SM , we will use the same symbol to identify all objects from $S_g M$ identified in SM .

3. Proof of Theorem 1.3

The goal of this section is to prove the following theorem.

Theorem 3.1. *Let M be a closed, connected, oriented surface, and let $\rho \in C^\infty(M, \mathbb{R}^+)$. Suppose (g, b_1) and $(\rho g, b_2)$ are two Anosov magnetic systems on M satisfying the following conditions:*

$$\text{MLS}(g, b_1) = \text{MLS}(\rho g, b_2), \quad \text{Area}(g) = \text{Area}(\rho g), \quad \text{and} \quad [b_1 \Omega_g] = [b_2 \Omega_g].$$

Then $\rho \equiv 1$ and $b_1 = b_2$.

Observe that [14, Theorem 1.2] reduces Theorem 1.3 to this setting. The proof that we present below follows, to some extent, the strategy of the proof in [31, Theorem 2] for Riemannian geodesics. In order to adapt the argument presented there to the magnetic case, we need to compare the magnetic length functional defined above with the Riemannian length functional. This comparison requires the lemma below, which allows us to bound the difference between the two functionals. Let $h : SM \rightarrow SM$ denote the conjugacy described in the Introduction on the level of SM .

Lemma 3.2. *Let M be a closed, connected, oriented surface. Suppose (g_1, b_1) and (g_2, b_2) are two Anosov magnetic systems on M satisfying the following properties:*

$$\text{MLS}(g_1, b_1) = \text{MLS}(g_2, b_2), \quad \text{Area}(g_1) = \text{Area}(g_2), \quad \text{and} \quad [b_1 \Omega_{g_1}] = [b_2 \Omega_{g_2}].$$

Then for every homologically trivial closed (g_1, b_1) -geodesic γ_1 , we have

$$\int_{\Sigma_{g_2, b_2}(\gamma_1)} b_2 \Omega_{g_2} = \int_{\gamma_1} \zeta, \tag{3.1}$$

where $b_2 \Omega_{g_2} - b_1 \Omega_{g_1} = d\zeta$.

Proof. As in (2.6), write $b_1 \Omega_{g_1} = c_{g_1, b_1} K^{g_1} \Omega_{g_1} + d\theta_1$. Define the 1-form

$$\kappa := -c_{g_1, b_1} \psi^{g_1} + p^*(\theta_1)$$

on SM , here using (2.3), and observe that we have $p^*(b_2 \Omega_{g_2}) = d(\kappa + p^*(\zeta))$. In particular, if γ_1 is a closed (g_1, b_1) -geodesic and $\tilde{\Sigma}_{g_2, b_2}(\gamma_1)$ is the lift of the 2-chain $\Sigma_{g_2, b_2}(\gamma_1)$ from M to SM , then Stokes' theorem yields

$$\begin{aligned} \int_{\Sigma_{g_2, b_2}(\gamma_1)} b_2 \Omega_{g_2} &= \int_{\tilde{\Sigma}_{g_2, b_2}(\gamma_1)} p^*(b_2 \Omega_{g_2}) \\ &= \left[\int_{\gamma_1} \kappa(X_{g_1, b_1}) - \int_{\gamma_2} \kappa(X_{g_2, b_2}) \right] + \left[\int_{\gamma_1} \zeta - \int_{\gamma_2} \zeta \right], \end{aligned}$$

where γ_2 is the image of γ_1 by h , i.e. the (g_2, b_2) -geodesic in the same free homotopy class as γ_1 . Using a change of variables,

$$\int_{\Sigma_{g_2, b_2}(\gamma_1)} b_2 \Omega_{g_2} = \int_{\gamma_2} [h_*(\kappa)(X_{g_2, b_2}) - \kappa(X_{g_2, b_2})] + \left[\int_{\gamma_1} \zeta - \int_{\gamma_2} \zeta \right]. \quad (3.2)$$

Next, using Lemma 2.1 and (2.5), we deduce that $\omega_{g_2, b_2} = h^*(\omega_{g_1, b_1})$. Since, by definition, $\omega_{g_i, b_i} = -d\alpha^{g_i} + p^*(b_i \Omega_{g_i})$, we observe that the 1-form $-\alpha^{g_2} + \kappa + p^*(\zeta) - h_*(-\alpha^{g_1} + \kappa)$ on SM is closed. Using the fact that h is a conjugacy, so $h_*(X_{g_1, b_1}) = X_{g_2, b_2}$, it follows that

$$\iota_{X_{g_2, b_2}}(-\alpha^{g_2} + \kappa + p^*(\zeta) - h_*(-\alpha^{g_1} + \kappa)) = \iota_{X_{g_2, b_2}}(\kappa - h_*(\kappa)) + \zeta,$$

where we now view ζ as a function on SM . Since γ_2 is also homologically trivial, the integral of the left hand side over the lift of γ_2 is zero, hence

$$0 = \int_{\gamma_2} [\kappa(X_{g_2, b_2}) - h_*(\kappa)(X_{g_2, b_2})] + \int_{\gamma_2} \zeta. \quad (3.3)$$

Substituting (3.3) into (3.2), the result follows. $\square$

Remark 3.3. We make a few observations.

- (1) In the case where the metrics are conformally equivalent, one can do this calculation directly on the unit tangent bundles without passing through the projective bundle. In particular, one can show that if γ is a unit g_1 -geodesic, $g_2 = \rho g_1$, and k^{g_i} is the geodesic curvature with respect to g_i , then there is a 1-form ζ on M such that

$$\int_{\gamma} k^{g_2} ds_{g_2} = \int_{\gamma} k^{g_1} ds_{g_1} + \int_{\gamma} \zeta.$$

- (2) Notice that if we have $[b_1 \Omega_{g_1}] = -[b_2 \Omega_{g_2}]$, so $b_2 \Omega_{g_2} + b_1 \Omega_{g_1} = d\zeta$, then (3.3) becomes

$$0 = - \int_{\gamma_2} [\kappa(X_{g_2, b_2}) + h^*(\kappa)(X_{g_2, b_2})] + \int_{\gamma_2} \zeta.$$

Substituting this into (3.2) yields

$$\int_{\Sigma_{g_2, b_2}(\gamma_1)} b_2 \Omega_{g_2} = -2 \int_{\gamma_2} \kappa(X_{g_2, b_2}) + \int_{\gamma_1} \zeta.$$

The term $\kappa(X_{g_2, b_2})$ presents a challenge in applying the upcoming ergodic arguments, and is why we need our cohomology class assumption.

With Lemma 3.2 in hand, we can now prove the result.

Proof of Theorem 3.1. For ease of notation, let $A := \text{Area}(g) = \text{Area}(\rho g)$, let ν^{g_i} denote the normalized Liouville volume measures, and let $g_1 := g$ and $g_2 := \rho g$. Jensen's inequality yields:

$$\int_{SM} \rho^{1/2} d\nu^{g_1} = \frac{1}{A} \int_M \rho^{1/2} \Omega_{g_1} \leq \left[\frac{1}{A} \int_M \rho \Omega_{g_1} \right]^{1/2} = 1,$$

with equality if and only if $\rho \equiv 1$. Let γ_1 be a unit speed (g_1, b_1) -geodesic, and let γ_2 be a unit speed (g_2, b_2) -geodesic in the same free homotopy class. Using the fact that magnetic geodesics minimize the magnetic length functional and the fact that $\ell_{g_i}(\gamma_i) = L_{g_i, b_i}(\gamma_i)$, we have

$$L_{g_1, b_1}(\gamma_1) = L_{g_2, b_2}(\gamma_2) \leq L_{g_2, b_2}(\gamma_1).$$

In particular, normalizing both sides by $\ell_{g_1}(\gamma_1)$ and using Lemma 3.2, we have

$$1 \leq \frac{1}{\ell_{g_1}(\gamma_1)} \int_{\gamma_1} \rho^{1/2} - \frac{1}{\ell_{g_1}(\gamma_1)} \int_{\gamma_1} \zeta.$$

Since this holds for every homologically trivial closed orbit, we can add weights to these terms and use Theorem 2.6 to deduce that

$$1 \leq \int_{SM} \rho^{1/2} d\nu^{g_1} - \int_{SM} \zeta d\nu^{g_1} \leq 1 - \int_{SM} \zeta d\nu^{g_1}.$$

Using flip invariance of the Liouville volume measure, the integral of ζ is zero. Thus, $\rho \equiv 1$.

Given $g_1 = g_2 =: g$, we now need to show that $b_1 = b_2$. Let γ_1 be a closed (g, b_1) -geodesic, and let γ_2 be the corresponding (g, b_2) -geodesic in the same free homotopy class. Lemma 2.5 yields

$$\ell_g(\gamma_2) = L_{g, b_2}(\gamma_2) \leq L_{g, b_2}(\gamma_1) = \ell_g(\gamma_1) - \int_{\Sigma_{g, b_2}(\gamma_1)} b_2 \Omega_g.$$

Consequently, the marked length spectrum assumption implies that for all closed (g, b_1) -geodesics, we have

$$\int_{\Sigma_{g,b_2}(\gamma_1)} b_2 \Omega_g \leq 0.$$

Now, recall that the Gauss-Bonnet theorem together with the fact that γ_1 and γ_2 are homotopic implies that

$$\int_{\Sigma_{g,b_2}(\gamma_1)} K^g \Omega_g + \int_{\gamma_1} k^g ds - \int_{\gamma_2} k^g ds = 0.$$

Since (g, b_i) -geodesics have g -geodesic curvature b_i thanks to (1.3), this can be reframed as

$$\int_{\Sigma_{g,b_2}(\gamma_1)} K^g \Omega_g = \int_{\gamma_2} b_2 - \int_{\gamma_1} b_1.$$

Recall from (2.6) that $b_2 \Omega_g = c_{g,b_2} K^g \Omega_g + d\theta_2$. Thus, using Stokes' theorem,

$$c_{g,b_2} \int_{\gamma_2} b_2 - c_{g,b_2} \int_{\gamma_1} b_1 + \int_{\gamma_1} \theta_2 - \int_{\gamma_2} \theta_2 \leq 0.$$

Note that the cohomology class assumption implies that $c_{g,b_1} = c_{g,b_2} =: c$. Also note that Lemma 2.4 implies that there is a closed 1-form ρ on M such that

$$-c \int_{\gamma_2} b_2 + \int_{\gamma_2} \theta_2 = -c \int_{\gamma_1} b_1 + \int_{\gamma_1} [\theta_1 + \rho],$$

Rearranging,

$$c \int_{\gamma_2} b_2 = c \int_{\gamma_1} b_1 - \int_{\gamma_1} [\theta_1 + \rho] + \int_{\gamma_2} \theta_2,$$

and substituting this in the above yields that for all closed (g, b_1) -geodesics,

$$\int_{\gamma_1} [\theta_2 - \theta_1 - \rho] \leq 0.$$

The non-positive Livshits theorem [34, Theorem 1] implies that there are Hölder continuous functions $F : S_g M \rightarrow [0, \infty)$ and $V : S_g M \rightarrow \mathbb{R}$ such that $\rho + \theta_1 - \theta_2 = X_{g_1, b_1}(V) + F$. Integrating this over $S_g M$ with respect to the normalized Liouville measure, we have

$$0 = \int_{S_g M} [\theta_2 - \theta_1 - \rho] d\nu^g = - \int_{S_g M} F d\nu^g.$$

Thus, $F \equiv 0$. Using [15, Theorem B], we see that $\theta_2 - \theta_1 - \rho$ is an exact 1-form on M , hence $\theta_2 - \theta_1$ is closed. Finally, observe that we have the relationship $(b_2 - b_1)\Omega_g = d(\theta_2 - \theta_1) = 0$ by (2.6), hence $b_2 = b_1$. $\square$

4. Examples

The goal of this section is to construct a series of examples in order to better understand the assumptions in Theorem 1.3, as well as to understand how Corollary 1.6 compares to known results on marked length spectrum rigidity. As a first step, we show how the area assumption cannot be removed.

Lemma 4.1. *Let M be a closed, connected, oriented surface, and let g be a hyperbolic metric on M . We have $\text{MLS}(g, b) = (1 - b^2)^{-1/2} \text{MLS}(g)$ for every constant $|b| < 1$.*

In particular, consider the homothetic metric $g_b := (1 - b^2)^{-1}g$. Lemma 4.1 shows that $\text{MLS}(g_b) = \text{MLS}(g, b)$ and $\text{Area}(g) \neq \text{Area}(g_b)$, and thus the area assumption in Theorem 1.3 is necessary.

Proof of Lemma 4.1. Lift the magnetic system to the upper-half plane $\mathbb{H}$. Denote the lifted magnetic system on $\mathbb{H}$ by $(\tilde{g}, b)$, where $\tilde{g}$ is given by $\tilde{g} = y^{-2}(dx^2 + dy^2)$. In particular, M can be viewed as the quotient of $\mathbb{H}$ under the action of a cocompact lattice Γ in $\text{PSL}(2, \mathbb{R})$.

Let $\nu \in \mathcal{C}$, and let $\gamma : \mathbb{R} \rightarrow M$ the unique closed g -geodesic representative. Without loss of generality, we assume that the lifted geodesic $\tilde{\gamma}$ is of the form

$$\tilde{\gamma}(\mathbb{R}) = \{(0, r) \in \mathbb{R}^2 \mid r > 0\}.$$

In particular, there exists an isometry $f \in \Gamma$ so that $f(\tilde{\gamma}(t)) = \tilde{\gamma}(t + \ell)$ for every $t \in \mathbb{R}$ (see, for example, [18]). Since f fixes $(0, 1)$ and the point at infinity, we deduce that $f(x, y) = e^{\ell_g(\gamma)}(x, y)$, thus

$$\ell_g(\gamma) = \int_1^{e^{\ell_g(\gamma)}} \frac{dy}{y}.$$

Let $\gamma_b : \mathbb{R} \rightarrow M$ be the unique closed (g, b) -geodesic with free homotopy class ν . Notice that we have

$$\tilde{\gamma}_b(\mathbb{R}) = \{(rb, r(1 - b^2)) \mid r > 0\}$$

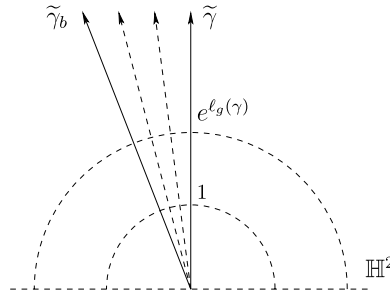


Fig. 1. An example showing how $\tilde{\gamma}$ moves as b changes.

and $f(b, 1 - b^2) = (e^{\ell_g(\gamma)}b, e^{\ell_g(\gamma)}(1 - b^2))$, hence

$$\ell_g(\gamma_b) = \int_1^{e^{\ell_g(\gamma)}} \frac{dy}{y\sqrt{1-b^2}} = \frac{\ell_g(\gamma)}{\sqrt{1-b^2}}.$$

(See Fig. 1.) The result follows $\square$

Next, we show how the assumption on the cohomology classes in Theorem 1.3 cannot be removed.

Lemma 4.2. *Let M be a closed, connected, oriented surface, and let g be a hyperbolic metric on M . We have $\text{MLS}(g, b) = \text{MLS}(g, -b)$ for every constant $|b| < 1$.*

Proof of Lemma 4.2. Define

$$X_{g,b} := \frac{1}{2} \begin{pmatrix} 1 & b \\ -b & -1 \end{pmatrix}.$$

It is well-known that there is a cocompact lattice Γ in $\text{PSL}(2, \mathbb{R})$ such that the magnetic flow associated to (g, b) on $S_g M$ can be interpreted as the flow on $\Gamma \backslash \text{PSL}(2, \mathbb{R})$ given by $\varphi_t^{g,b}(\Gamma q) := \Gamma q e^{tX_b}$ (see, for example, [41]). Observe that the matrix $c \in \text{SL}(2, \mathbb{R})$ given by

$$c := \frac{1}{1-b^2} \begin{pmatrix} 1 & -b \\ -b & 1 \end{pmatrix}$$

satisfies the condition that $X_{g,b}c = cX_{g,-b}$. Thus, an easy argument using induction and the definition of the exponential map tells us that we have $h \circ \varphi_{g,b}^t = \varphi_{g,-b}^t \circ h$ for the map $h : \Gamma \backslash \text{PSL}(2, \mathbb{R}) \rightarrow \Gamma \backslash \text{PSL}(2, \mathbb{R})$ given by $h(\Gamma q) := \Gamma qc$. This is a smooth conjugacy between the magnetic flows for (g, b) and $(g, -b)$. Furthermore, if we define a family of matrices $c_s \in \text{SL}(2, \mathbb{R})$ by

$$c_s := \frac{1}{1 - sb^2} \begin{pmatrix} 1 & -sb \\ -sb & 1 \end{pmatrix},$$

then we have an associated family of maps $h_s : \Gamma \backslash \mathrm{PSL}(2, \mathbb{R}) \rightarrow \Gamma \backslash \mathrm{PSL}(2, \mathbb{R})$ given by $h_s(\Gamma q) := \Gamma q c_s$. Observe that the family is continuous in s and satisfies $h_0 = \mathrm{Id}$ and $h_1 = h$, so h is homotopic to the identity. As discussed in [7, Section 10.3] and [11, Section 3], the subgroup of $\pi_1(SM)$ generated by the kernel of the footprint map equals the center of $\pi_1(SM)$. Since h is invertible, we have that h_* is an isomorphism on the level of the fundamental groups, hence induces an isomorphism on the level of $\pi_1(M)$. This shows that $\mathrm{MLS}(g, b) = \mathrm{MLS}(g, -b)$. $\square$

Finally, we construct an example of a closed, oriented, connected surface such that there exists an exact Anosov magnetic system (g, b) where the metric g is not Anosov by modifying the arguments in [8, Section 7]. This shows that it is not necessary to assume that g is Anosov in Corollary 1.6. We recall the following lemma, and briefly sketch its proof. For more details, we point the reader to [8, Sections 6 and 7].

Lemma 4.3. *There exists a closed, connected, orientable surface M and an Anosov magnetic system (g, b) on M with b constant such that g is not Anosov.*

Sketch of Proof. We will need the following.

Lemma 4.4 ([8, Lemma 6.1]). *Let (g, b) be a magnetic system on M . If there are constants $T > 0$ and $H > 1$ such that for any (g, b) -geodesic γ and any solution u to the magnetic Riccati equation*

$$\dot{u}(t) + u^2(t) + K^{g,b}(\gamma(t), \dot{\gamma}(t)) = 0 \quad (4.1)$$

with $u(0) \geq 0$, we have $1/H \leq u(T) \leq H$, then the magnetic system is Anosov.

Consider a rotationally symmetric surface $N = \mathbb{R} \times S^1$ with coordinates (s, θ) . For a smooth function $u : \mathbb{R} \rightarrow \mathbb{R}$, equip N with the following Riemannian metric:

$$g_u = ds^2 + \exp \left(\int_0^s u(\tau) d\tau \right)^2 d\theta^2.$$

Note that the Gaussian curvature is given by the formula

$$K^{g_u}(s, \theta) = -u'(s) - u^2(s),$$

where $'$ denotes the derivative with respect to the s coordinate. For $s \in \mathbb{R}$ the unit speed parallel

$$\alpha_s^\pm(t) = \left(s, \pm \exp \left(\int_0^s u(\tau) d\tau \right)^{-1} t \right)$$

has g_u -geodesic curvature $k^{g_u}(t) = \pm u(s)$. In particular, $\alpha_s^\pm$ is a closed g_u -geodesic if and only if $u(s) = 0$ [8, Lemma 6.2]. For $\delta > 0$ sufficiently small, consider the interval $[-\delta, \delta] \subset [-1/4, 1/4]$. Let $u(s) = \tanh(s)$, except for a small C^0 perturbation in the interval $[-\delta, \delta]$ where the function u satisfies the following:

- $|u(s)| < \tanh(1/4)$ and $K^{g_u}(s) < 1/4$, $\forall s \in [-\delta, \delta]$.
- $u(0) = 0$ and $u'(0) < 0$.

With this choice of u , we can immediately deduce that $\alpha_0^\pm(t) = (0, t)$ is a closed g_u -geodesic and $K^{g_u}(\gamma_0(t)) > 0$.

Denote by $(M, \bar{g}_u)$ the Riemannian surface obtained through a standard compactification of (N, g_u) – see, for instance, [5]. By construction, the metric $\bar{g}_u$ carries a closed geodesic contained in a region with positive curvature, hence the geodesic flow cannot be Anosov. Consider now the magnetic system $(g_u, 1/2)$ on N , and observe that the magnetic curvature is given by $K^{g_u, 1/2}(s) = K^{g_u}(s) + 1/4$. In particular, note that $K^{g_u, 1/2}$ is at most $1/2$ in the region $C_\delta = [-\delta, \delta] \times S^1$, and at most $-3/4$ outside C_δ . Observe also that, due to the choice of u , for $s \in [-1/4, 1/4]$ the g_u -geodesic curvature of a parallel is bounded in absolute value by $1/4$ and unit speed $(g_u, 1/2)$ -geodesics have prescribed g_u -geodesic curvature equal to $1/2$. Thus, each unit speed $(g_u, 1/2)$ -geodesic connected segment in the region $C_{1/4} = [-1/4, 1/4] \times S^1$ can be tangent to a parallel $\alpha_s^\pm$ at most once, and cannot be asymptotic to $\alpha_s^\pm$. Moreover, [8, Lemma 7.1] establishes that the amount of time unit speed $(g_u, 1/2)$ -geodesics spend in C_δ is bounded by $2t_\delta$ for a constant $t_\delta > 0$. Therefore, every $(\bar{g}_u, 1/2)$ -geodesic on M is contained in a region with negative magnetic curvature equal to $-3/4$, except on intervals with length bounded by $2t_\delta$ where the curvature is bounded by $1/2$. Consider the *magnetic Jacobi equation* along a $(\bar{g}_u, 1/2)$ -geodesic γ :

$$\ddot{y} + K^{g_u, 1/2} y = 0. \quad (4.2)$$

Functions of the form $u(t) = \dot{y}(t)/y(t)$, where $y(t)$ is the unique solution of (4.2) with initial condition $y(0) \neq 0$, are examples of solutions to (4.1) along γ . It is easy to check that for this choice of parameters, the assumptions of Lemma 4.4 are satisfied for $T = 1/8$ and $H > 0$ depending on t_δ , hence the magnetic flow associated to $(\bar{g}_u, 1/2)$ is Anosov. $\square$

The goal is to now modify this argument in order to derive the following.

Lemma 4.5. *There exists a closed, connected, orientable surface M and an exact Anosov magnetic system (g, b) on M such that g is not Anosov.*

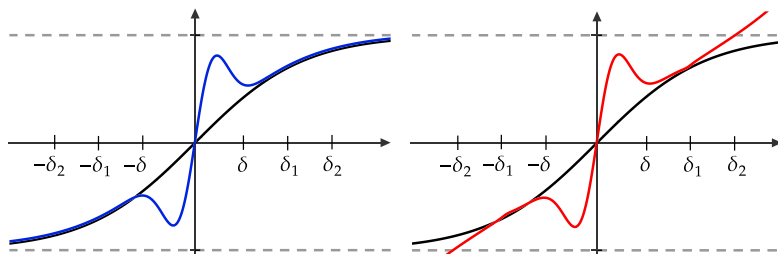


Fig. 2. The graphs of the functions u and v , each sketched against the graph of $\tanh$.

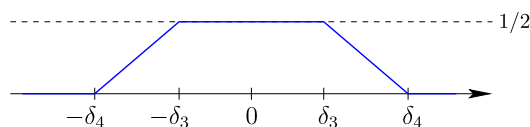


Fig. 3. An example of b_+ .

We briefly sketch the idea of the proof. The first step is to modify Lemma 4.3 in order to allow a magnetic intensity b_+ which is a small positive bump function. In particular, this will show that any (g, b_+) -geodesic must spend a small amount of time in the “bad region,” and we can shrink this amount of time as necessary. A symmetric argument will also show that one can use the magnetic intensity $b_- = -b_+$, and any (g, b_-) geodesic will also spend the same amount of time in the “bad region.” By gluing together these examples and taking the magnetic intensity $b := b_+ + b_-$ (note that these are supported on two disjoint regions), we see that any magnetic geodesic can only spend a small amount of time in the “bad regions,” and adjusting parameters can shrink this amount of time to be arbitrarily small. As long as we take T sufficiently large depending on the distance between the “bad regions,” as well as their size, then we can find an H where Lemma 4.4 applies. Moreover, since these “bad regions,” have the same size, we see that the average of b with respect to the area is zero, and hence this gives an example of an exact Anosov magnetic system whose underlying metric is not Anosov. We now present the details.

Proof of Lemma 4.5. We use the same notation as in the sketch of Lemma 4.3. The goal is to construct a magnetic intensity which is compactly supported on N , and whose growth is small. This will allow us to control the magnetic curvature as in the previous argument. Fix $\epsilon > 0$ sufficiently small, and let $\delta_2 > \delta_1 \gg 1/4$. Let v be a C^0 perturbation of u such that v coincides with u on the interval $[-\delta_1, \delta_1]$, $K^{g_v}(s) \leq 1$ for $|\delta_1| \leq |s| \leq |\delta_2|$ and $K^{g_v}(s) = -1 - \epsilon$ for $|s| > \delta_2$. (See Fig. 2.)

Let $\delta_4 \gg \delta_3 \gg \delta_2$. Let b_+ be a bump function with the following properties:

- b_+ is supported in $[-\delta_4, \delta_4]$
- $0 \leq b_+ \leq 1/2$ and $b_+ \equiv 1/2$ on $[-\delta_2, \delta_2]$,
- $\|db_+\|_\infty + b_+^2 < 1/4 + \epsilon$. (See Fig. 3.)

The magnetic curvature of (g_v, b_+) satisfies $K^{g,b_+}(s) = K^{g,1/2} \leq 1/2$ for $s \in [-\delta, \delta]$, and $K^{g,b_+}(s) \leq -3/4$ otherwise. By construction, the dynamics of (g_v, b_+) restricted on $C_{1/4}$ coincides with the dynamics of $(g, 1/2)$. Involving the same argument as in Lemma 4.3, we can conclude that the time any (g_v, b_+) -geodesic spends in C_δ is bounded above by $2t_\delta$ for a constant $t_\delta > 0$ depending on the parameters. Let $b_- := -b_+$. By symmetry, the time any (g_v, b_-) -geodesic spends in C_δ is bounded above by $2t_\delta$ for the same constant $t_\delta > 0$. For both of these examples, perform the same compactification procedure to get a closed, connected, oriented surface, and two magnetic systems $(\bar{g}_v, b_\pm)$.

Away from the neighborhoods C_δ , we take two open sets and glue together two copies of $(M, \bar{g}_v)$ along these open sets to get $\bar{M} = [M \cup M]_{g_v}$. Equip this new surface $\bar{M}$ with the magnetic system $(\bar{g}_v, b)$, where $b := b_+ + b_-$. By construction, the time each unit speed (g, b) -geodesic connected segment spends in the region corresponding through the glue to C_δ is still bounded by t_δ . Outside those regions, the magnetic curvature is negative and still bounded by $-3/4$, and thus by taking T sufficiently large depending on the distance between the regions C_δ , and taking H depending on t_δ and this distance, one can conclude from Lemma 4.4 that the corresponding flow is Anosov. Moreover, we have

$$\int_{\bar{M}} b \Omega_g = 2\pi \left[\int_{-\delta_4}^{\delta_4} b_+ ds + \int_{-\delta_4}^{\delta_4} b_- ds \right] = 0,$$

hence the system is exact. $\square$

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